

Room Criterion (RC) Mark II ratings for the octave band, unweighted, LEQ spectrum of the measurement file.

Settings. There are no user settings available for this mode.

PSD mode. PSD mode analyses the overall power spectral density using a variation of the Welch Modified Periodogram method.

Settings. 1. Select PSD→FFT Size from the main menu (1) [Fig. 6], to select the FFT size from the 256, 512, 1024, 2048, 4096 or 8192 pt FFT options.

2. Select PSD→Overlap from the main menu (1) [Fig. 6], to select the overlap percentage.

3. Select PSD→Window from the main menu (1) [Fig. 6], to select the FFT window type (Hanning, Hamming or Flat Top).

Sgram mode. The Sgram button breaks the measurement file into small segments of a selectable size and computes the Welch Modified Periodogram method for each segment.

Settings. 1. Select Spectrogram→FFT Size from the main menu (1) [Fig. 7], to select the FFT size from the

256, 512, 1024, 2048, or 4096 pt FFT options (default is 512).

2. Select Spectrogram→Slice Length from the main menu (1) [Fig. 7], to select the size of the buffer for each segment from the options of 100ms, 1s and 10s.

3. Select Spectrogram→Plot Scale from the main menu (1) [Fig. 7], to set the maximum dynamic range in each plot. The software truncates all data above 30 dB and 50 dB when 30 dB and 50 dB are selected, respectively. Selecting *Auto* will let the software choose the plot scale.

4. Select Spectrogram→View from the main menu (1) [Fig. 7], to display the resulting time-frequency plot (as 2D or 3D plots).

Save Analysis. This button saves the most recent analysis to a text file.

Different data are saved for different analysis modes:

1. For Lp plot mode, the time and Lp are saved.

2. For LEQ mode, the time and LEQ are saved. The header will include Ln statistics and Noise Dose information

(if applicable).

3. For Band mode, the centre frequencies and Band LEQ are saved.

4. For NC/RC mode, the NC/RC levels and octave band LEQ are saved.

5. For PSD mode, the frequency and power spectral density are saved.

6. Sgram mode has too much data to be saved in a text file, so the display figure is copied as a MATLAB figure and figure export function is called.

Save Plot

This button copies the current display figure as a MATLAB figure, and then the export function is called, which can be used to alter the size and font of the figure and then save it.

If you have been looking for a sound level meter to analyse the sound output from your audio device, this tool is for you. You can use it to play around with a calibrated sound recording, and implement different analysis techniques and observe their effect on the signals. Have fun! ●

The author is a technical journalist at EFY Gurgaon

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